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Day 3

Aim: To perform targeted TCP port scanning and a small authorized UDP scan using Nmap, identify at least three running services, and capture and analyze a relevant packet flow using Wireshark.

Objective:

- To perform targeted TCP scans using Nmap.
- To perform a small authorized UDP scan.
- To identify at least three services running on the target machine.
- To capture network packets generated during the scan using Wireshark.

Requirements:

Operating System: Kali and Windows

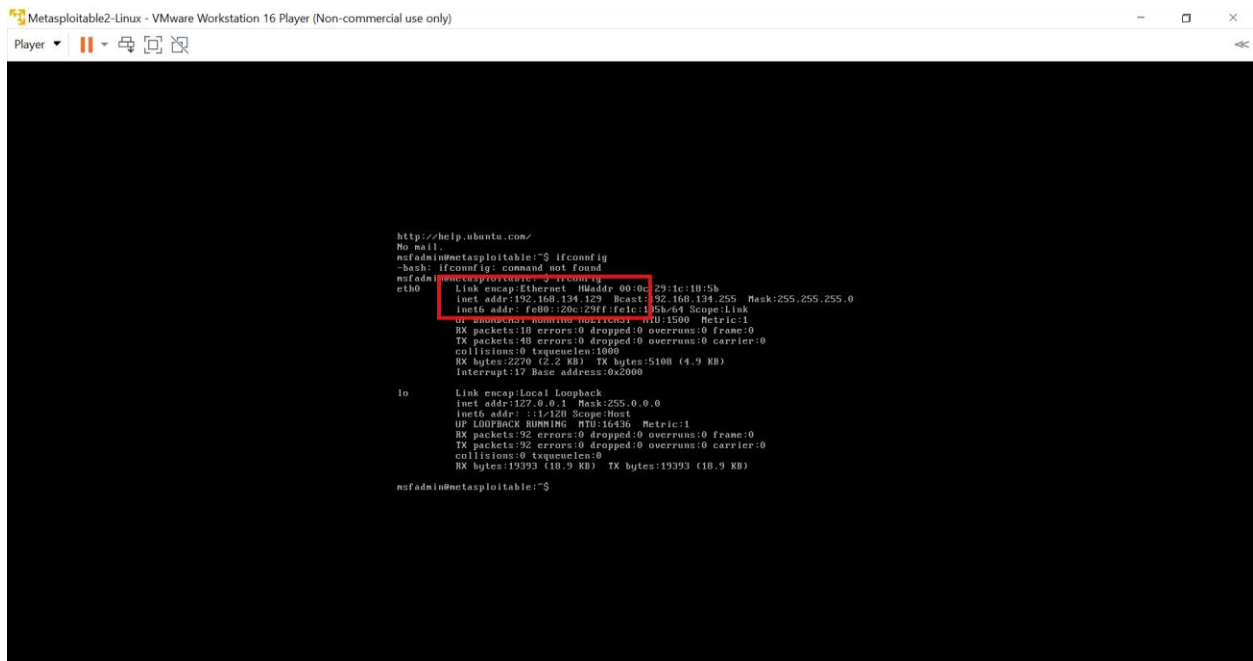
Tools: Nmap & Wireshark

Theory:

Nmap is a network scanning tool used to identify open ports and running services on a target system. A TCP SYN scan (-sS) checks selected TCP ports by analyzing SYN responses. A UDP scan (-sU) checks selected UDP ports and identifies their status. Wireshark is a packet analyzer used to capture and examine network traffic, such as the SYN → SYN/ACK → RST flow during a TCP scan.

Practical:

Step 1. Check the Metasploitable IP and kali ip address using ifconfig.

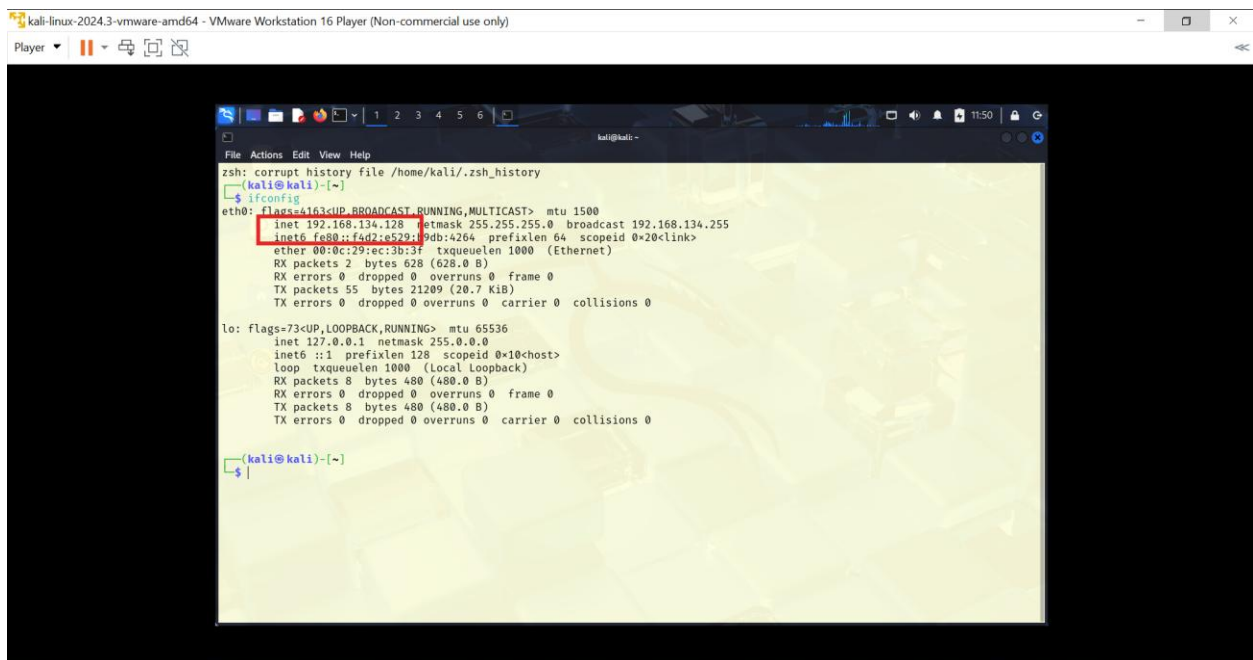


```
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
-bash: ifconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.134.129 netmask 255.255.255.0 broadcast 192.168.134.255
    inet6 fe80::20c:29:ec:3b:3f prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets 2 bytes 628 (628.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55 bytes 21209 (20.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msfadmin@metasploitable:~$
```

Figure 1. Metasploitable ip: 192.168.134.129



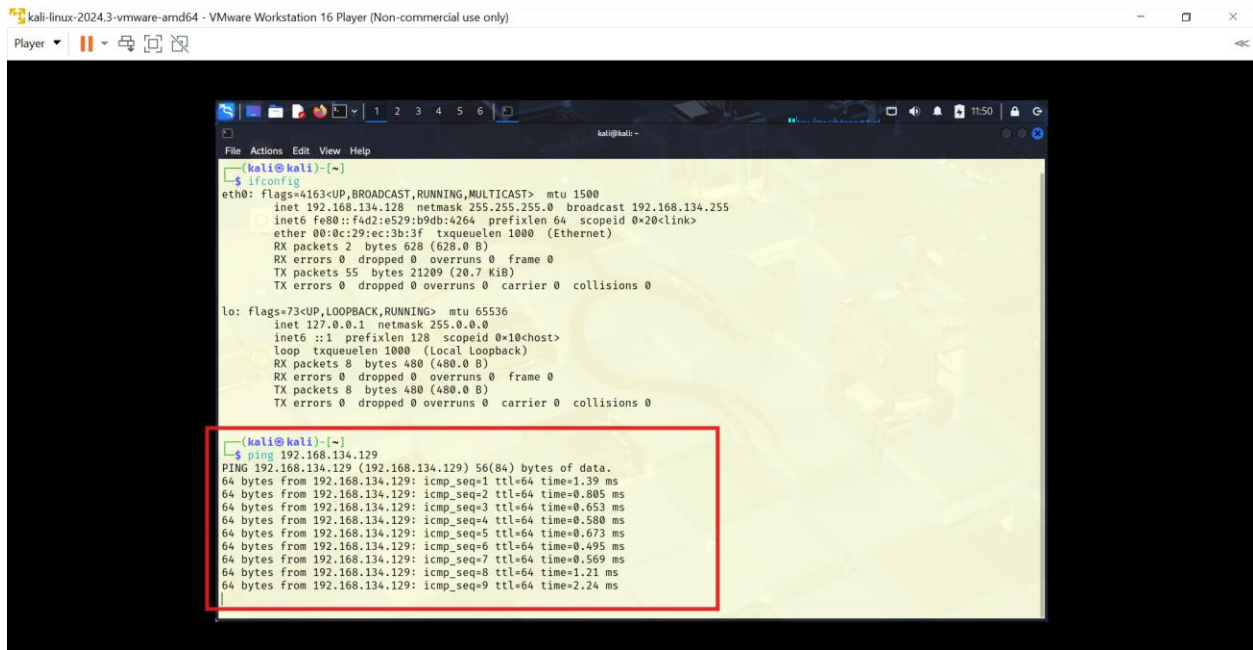
```
kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.134.128 netmask 255.255.255.0 broadcast 192.168.134.255
    inet6 fe80::20c:29:ec:3b:3f prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets 2 bytes 628 (628.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55 bytes 21209 (20.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

kali@kali:~$
```

Figure 2. Kali ip: 192.168.134.128

Step 2. Check connectivity using ping command.



```
kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.134.128 netmask 255.255.255.0 broadcast 192.168.134.255
    inet6 fe80::f4d2:e529:b9db:4264 prefixlen 64 scopeid 0<link>
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets 2 bytes 628 (628.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55 bytes 21209 (20.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

kali@kali:~$ ping 192.168.134.129
PING 192.168.134.129 (192.168.134.129) 56(84) bytes of data.
64 bytes from 192.168.134.129: icmp_seq=1 ttl=64 time=1.39 ms
64 bytes from 192.168.134.129: icmp_seq=2 ttl=64 time=0.805 ms
64 bytes from 192.168.134.129: icmp_seq=3 ttl=64 time=0.653 ms
64 bytes from 192.168.134.129: icmp_seq=4 ttl=64 time=0.580 ms
64 bytes from 192.168.134.129: icmp_seq=5 ttl=64 time=0.673 ms
64 bytes from 192.168.134.129: icmp_seq=6 ttl=64 time=0.495 ms
64 bytes from 192.168.134.129: icmp_seq=7 ttl=64 time=0.569 ms
64 bytes from 192.168.134.129: icmp_seq=8 ttl=64 time=1.21 ms
64 bytes from 192.168.134.129: icmp_seq=9 ttl=64 time=2.24 ms
```

Figure 3. ping 192.168.134.129

Step 3. Open wireshark using command like sudo wireshark..

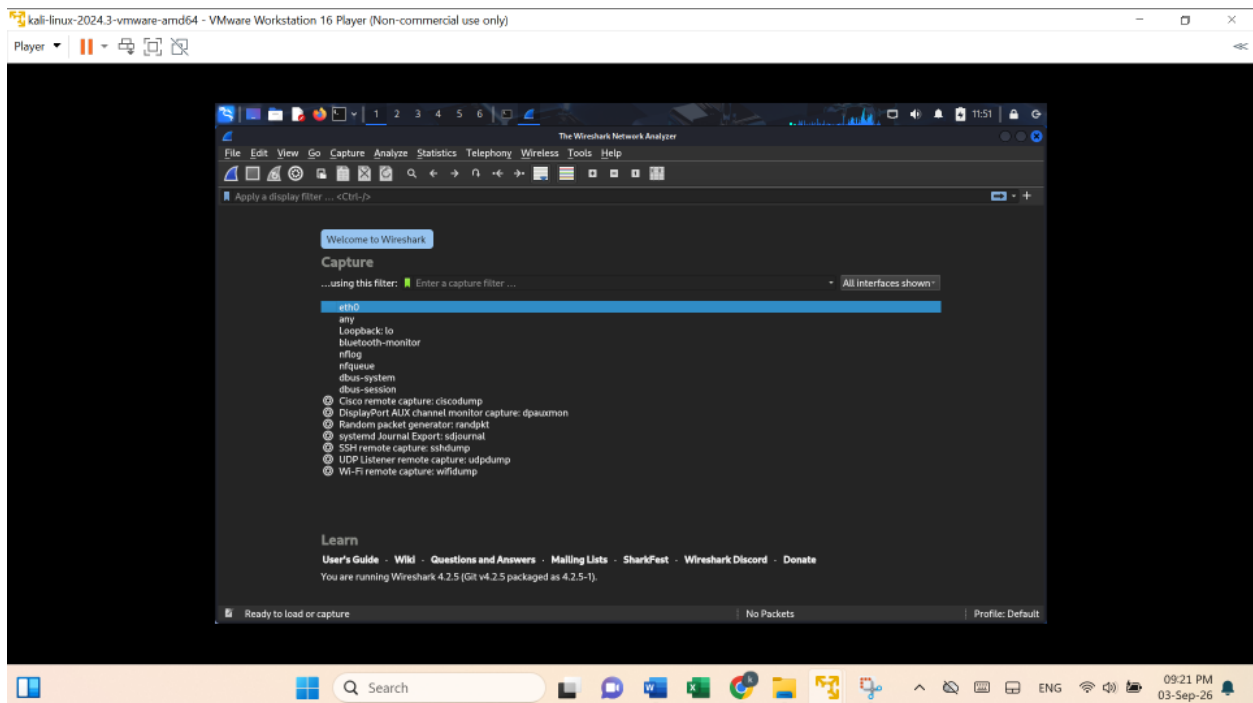


Figure 4. In wireshark, Click on eth0

Step 4. Capture packets, using different commands like related to tcp anf udp scanning..

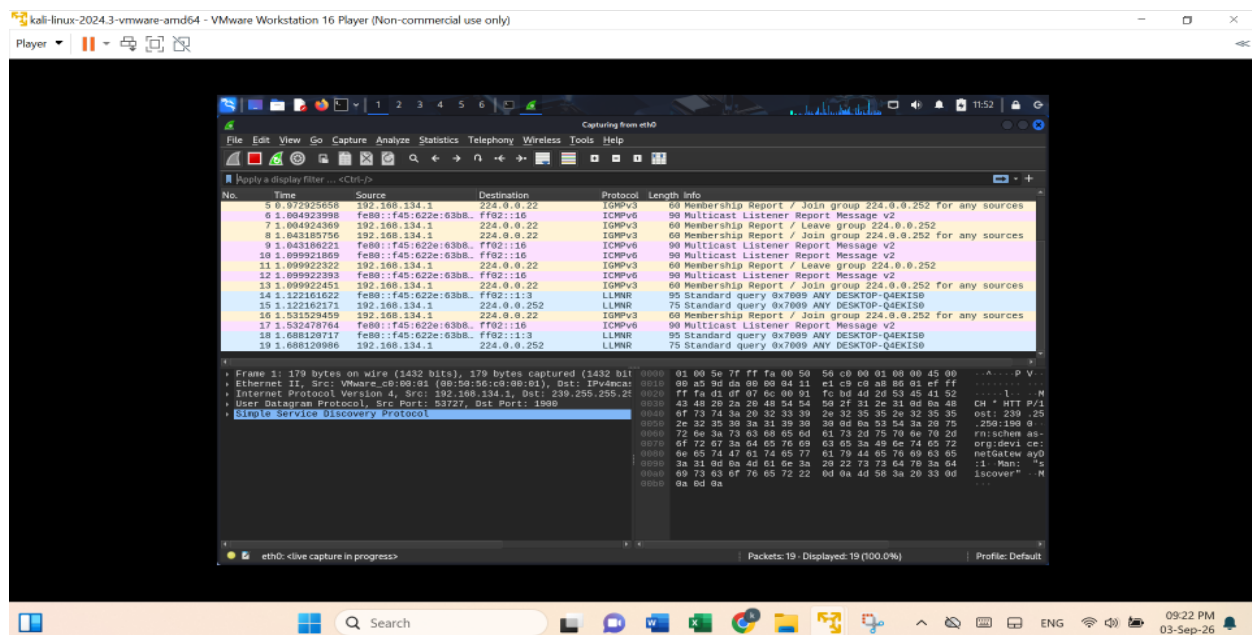


Figure 5. Packet capturing...

Step 5. Perform a targeted TCP SYN scan on the Metasploitable target using Nmap and identify open services.

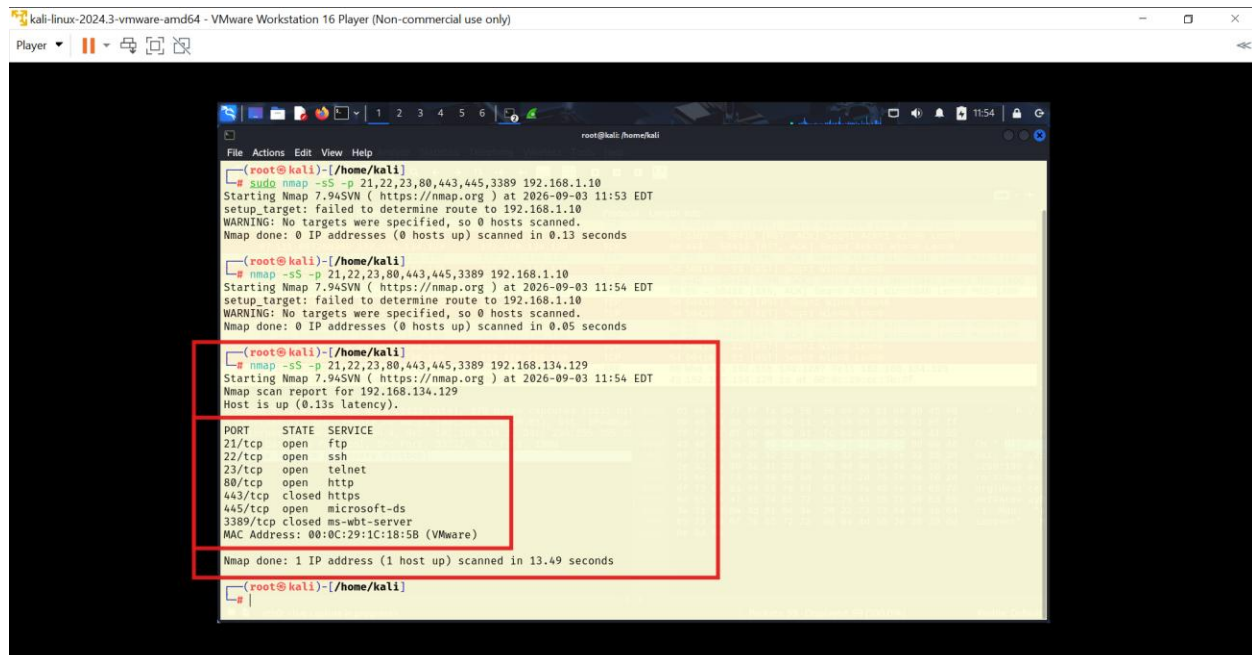
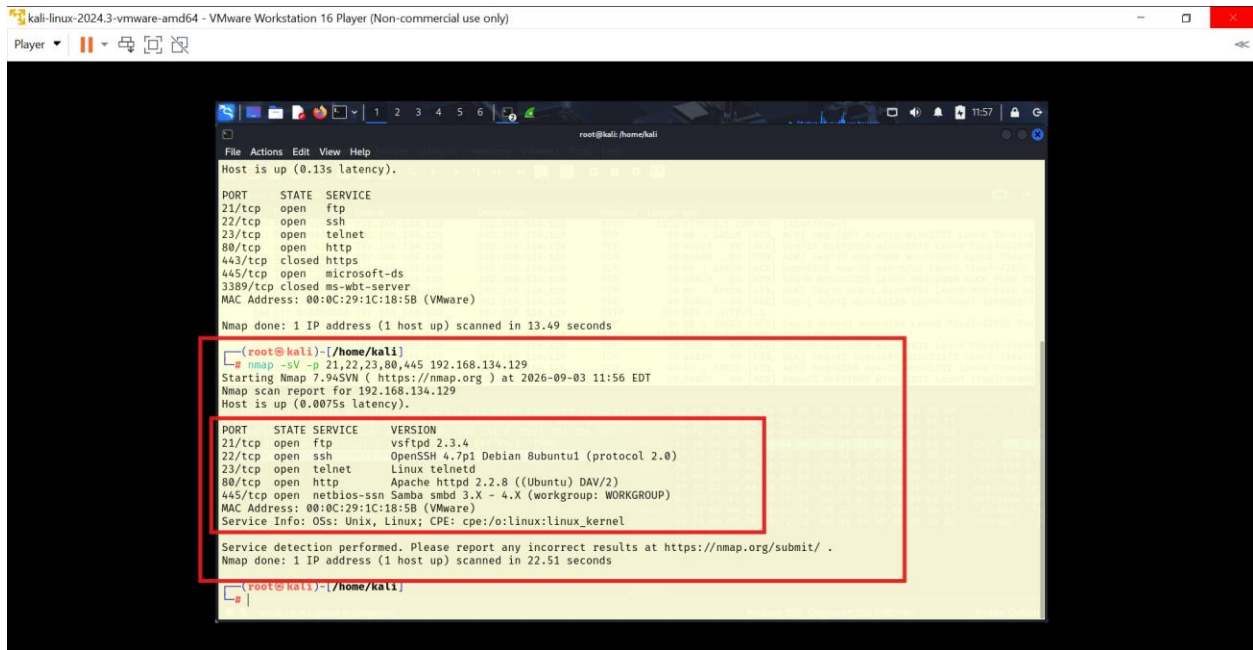


Figure 6. nmap -sS -p 21,22,23,80,443,3389 192.168.134.129

Step 6. Perform service version detection using Nmap to identify the versions of the open services.



```
kali-linux-2024.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
root@kali: /home/kali

Host is up (0.13s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
80/tcp    open  http
443/tcp   closed https
445/tcp    open  microsoft-ds
3389/tcp  closed ms-wbt-server
MAC Address: 00:0C:29:1C:18:5B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.49 seconds

(root@kali)~[/home/kali]
# nmap -sV -p 21,22,23,80,445 192.168.134.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-03 11:56 EDT
Nmap scan report for 192.168.134.129
Host is up (0.0075s latency).

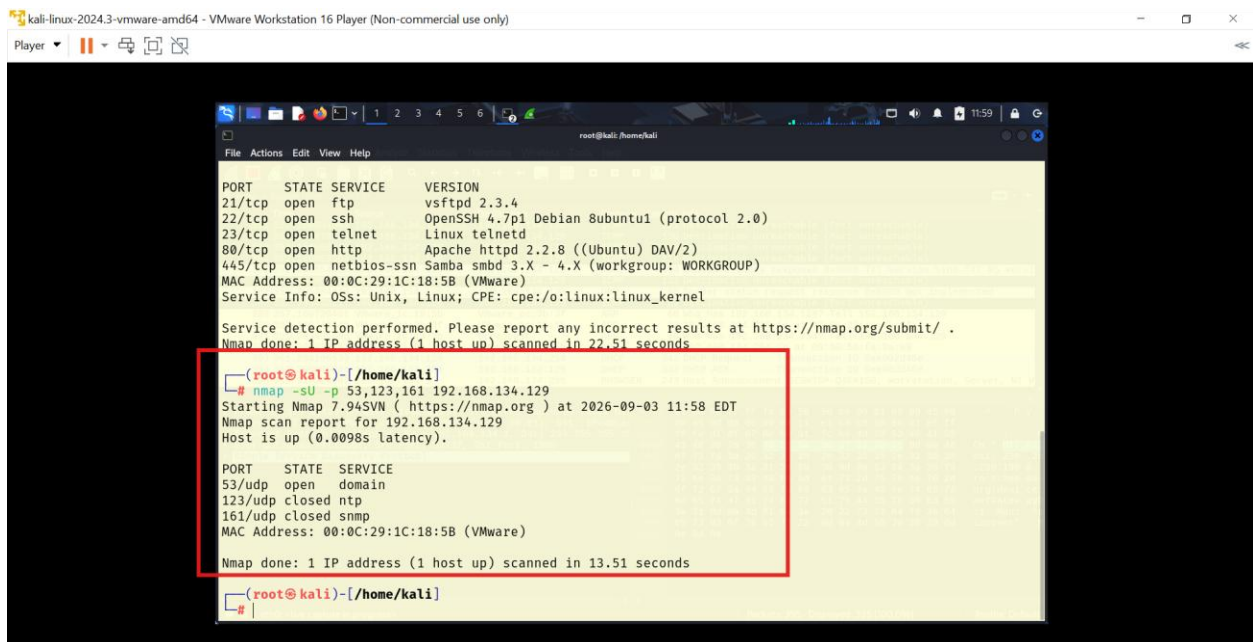
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:1C:18:5B (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.51 seconds

(root@kali)~[/home/kali]
#
```

Figure 7. `nmap -sV -p 21,22,23,80,445 192.168.134.129`

Step 7. Perform a small UDP scan using Nmap and identify the status of the selected UDP ports.



```
kali-linux-2024.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
root@kali: /home/kali

PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:1C:18:5B (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.51 seconds

(root@kali)~[/home/kali]
# nmap -sU -p 53,123,161 192.168.134.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-03 11:58 EDT
Nmap scan report for 192.168.134.129
Host is up (0.0098s latency).

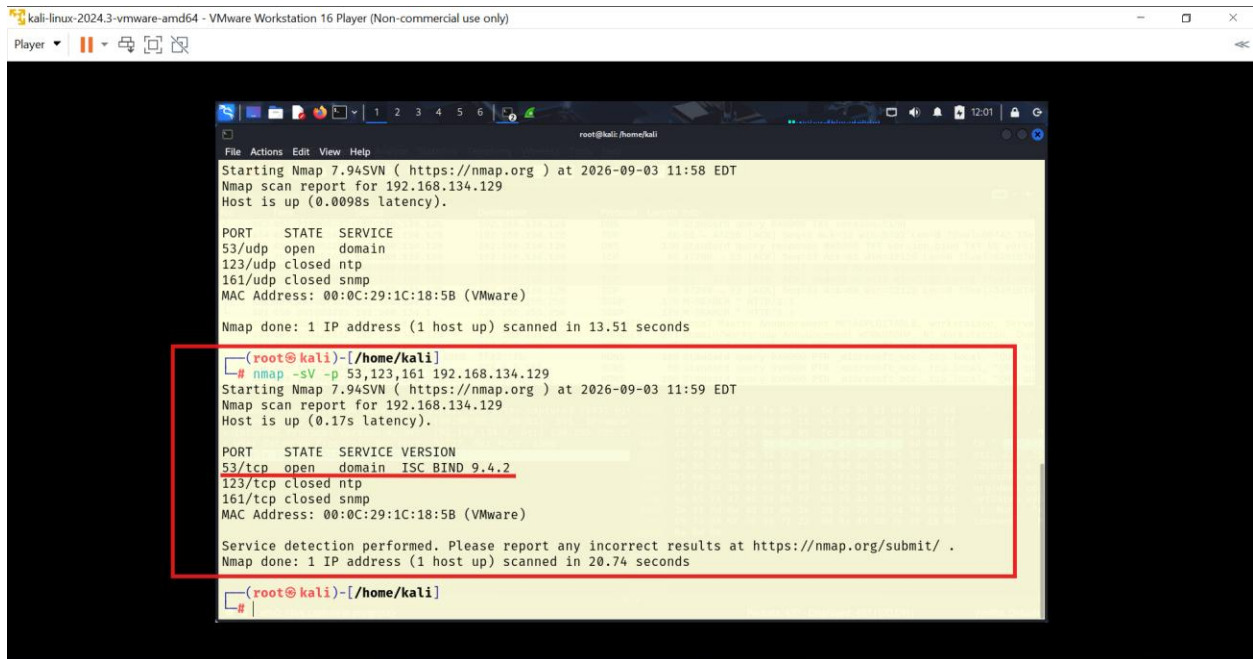
PORT      STATE SERVICE
53/udp    open  domain
123/udp   closed ntp
161/udp   closed snmp
MAC Address: 00:0C:29:1C:18:5B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.51 seconds

(root@kali)~[/home/kali]
#
```

Figure 8. `Nmap -sU -p 53,123,161 192.168.134.129`

Step 8. Perform UDP service version detection using Nmap to identify the version of the open UDP service.



```
root@kali:~/home/kali
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-03 11:58 EDT
Nmap scan report for 192.168.134.129
Host is up (0.0098s latency).

PORT      STATE SERVICE
53/udp    open  domain
123/udp   closed ntp
161/udp   closed snmp
MAC Address: 00:0C:29:1C:18:5B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 13.51 seconds

(root@kali)~[/home/kali]
# nmap -sV -p 53,123,161 192.168.134.129
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-09-03 11:59 EDT
Nmap scan report for 192.168.134.129
Host is up (0.17s latency).

PORT      STATE SERVICE VERSION
53/tcp    open  domain  ISC BIND 9.4.2
123/tcp   closed ntp
161/tcp   closed snmp
MAC Address: 00:0C:29:1C:18:5B (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 20.74 seconds

(root@kali)~[/home/kali]
#
```

Figure 9. Nmap -sV -p 53,123,161 192.168.134.129

Step 9. Save it, name like internship.pcap..

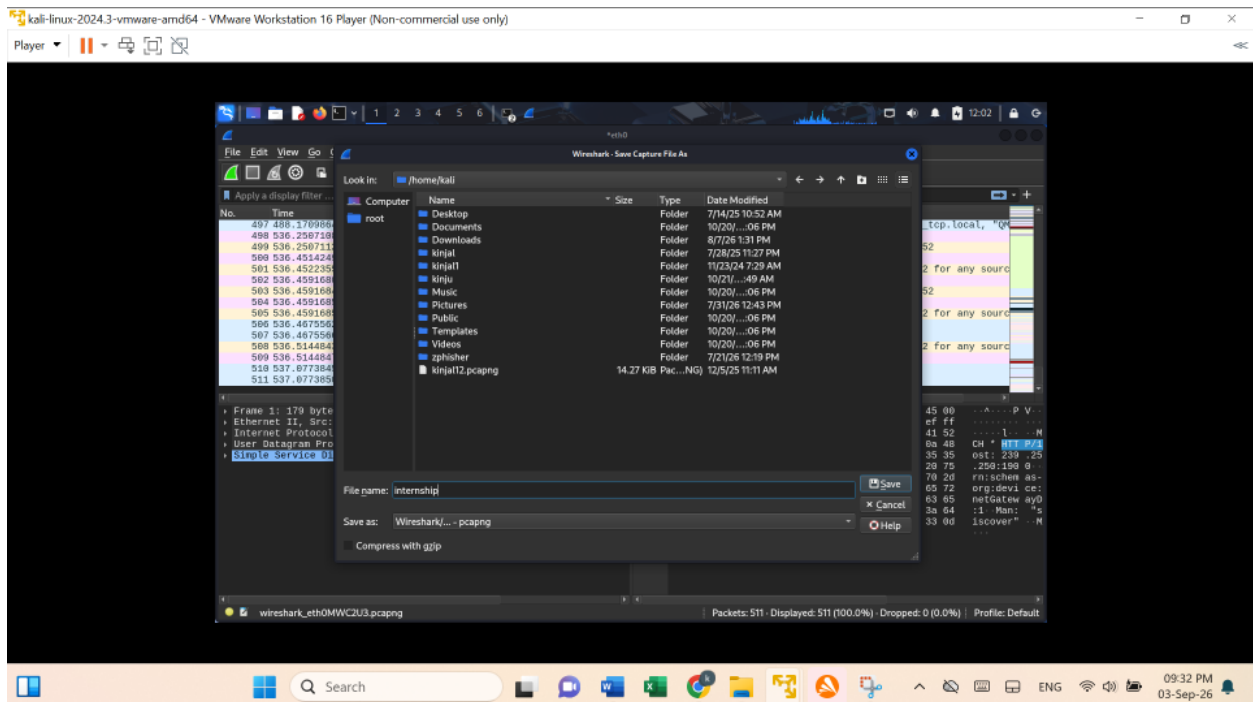


Figure 10. Save it..

Step 10. Capture and analyze the TCP SYN packet flow in Wireshark using the target IP filter.

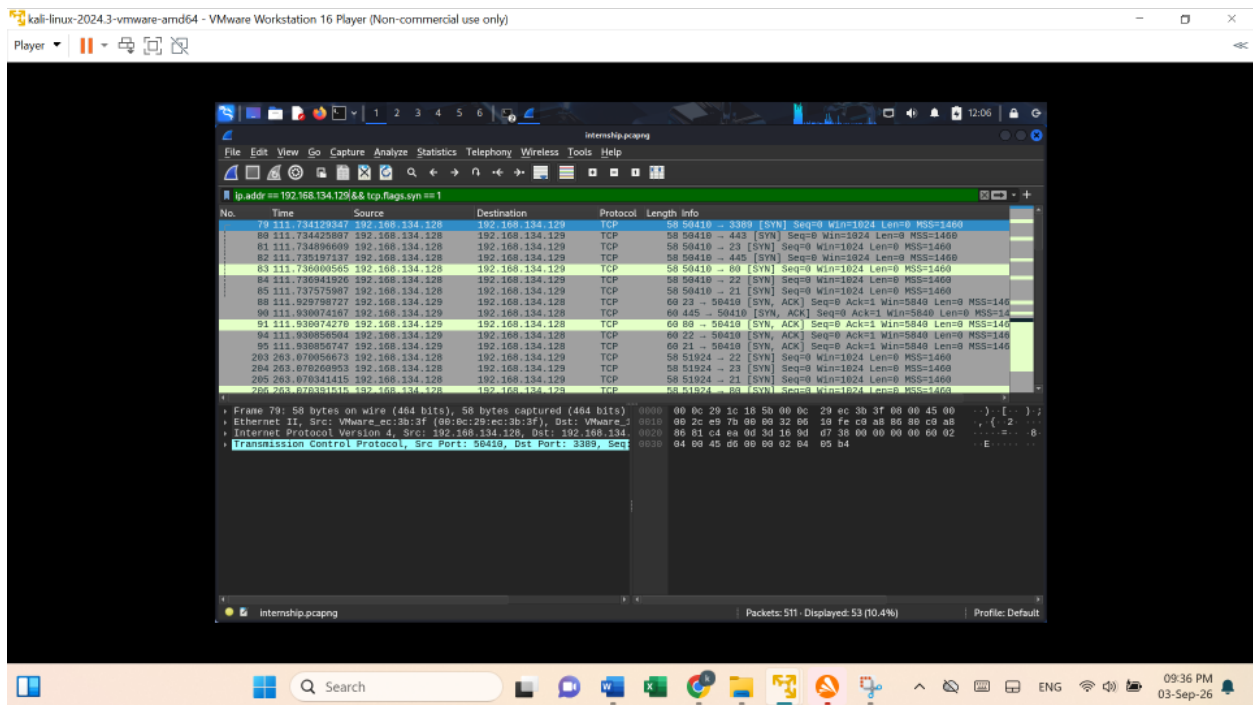


Figure 11. `ip.addr == 192.168.134.129 && tcp.flags.syn == 1`

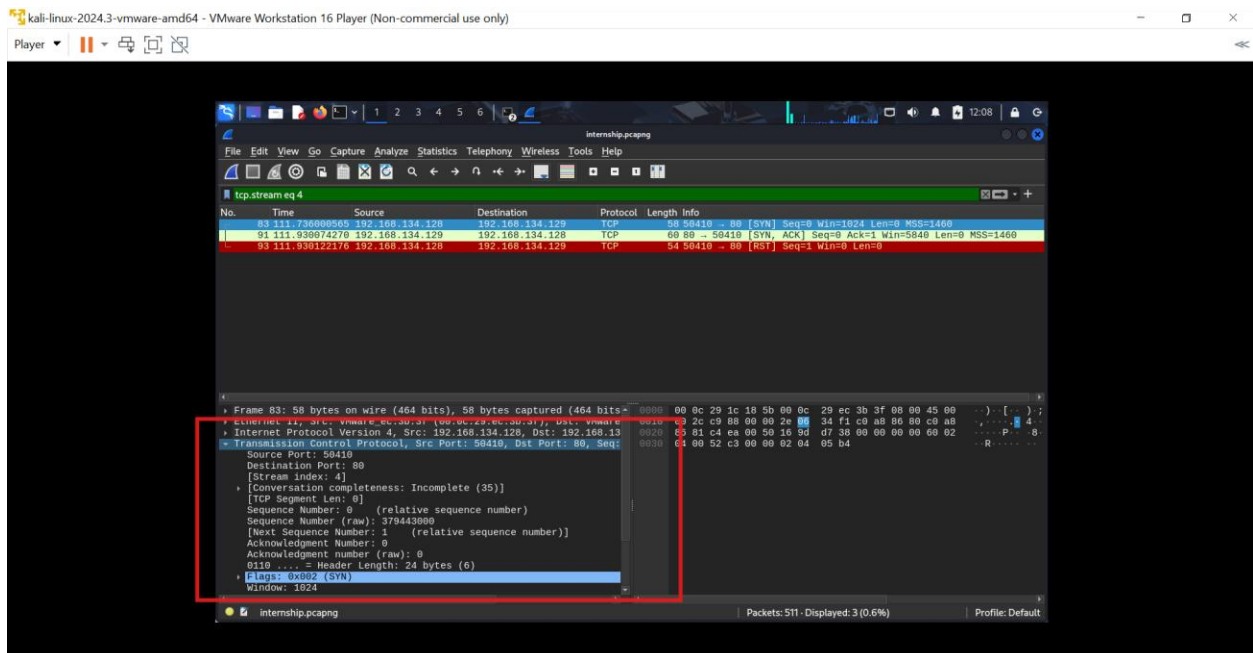


Figure 12. Check it..

Step 11. Analyze the captured UDP packets and ICMP responses in Wireshark using the target IP filter.

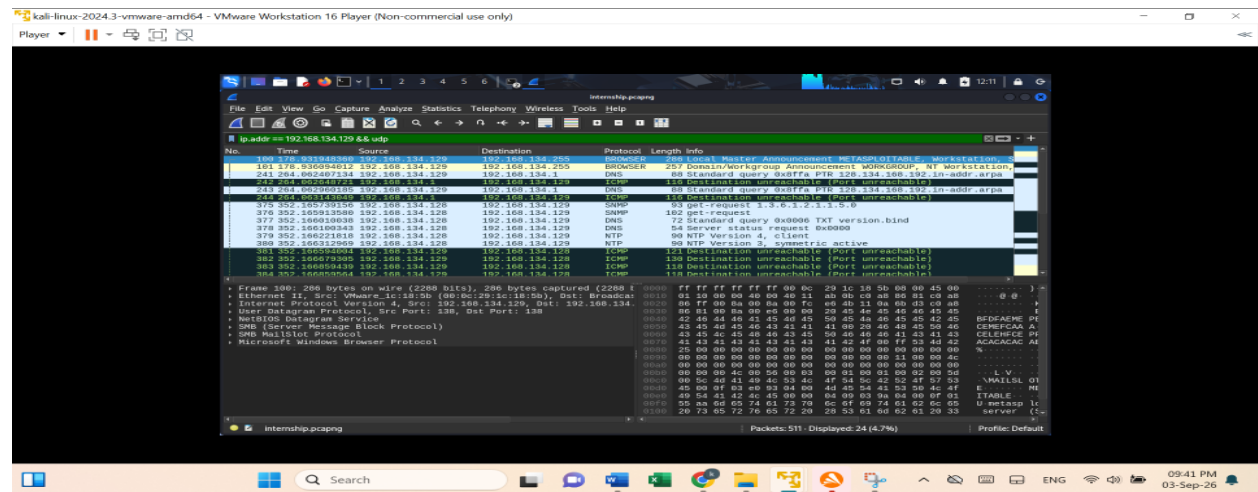


Figure 13. `ip.addr == 192.168.134.129 && udp`

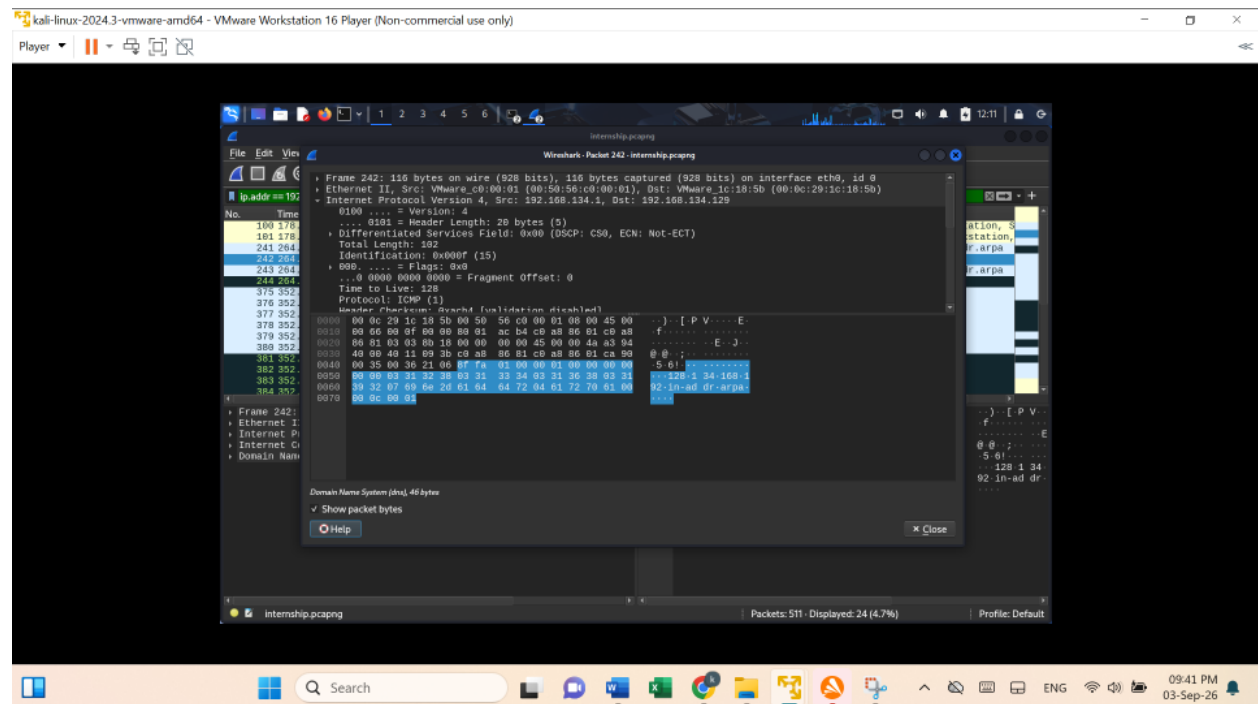


Figure 14. Check it.

Conclusion:

I successfully performed targeted TCP and small UDP scans on the authorized target using Nmap. I identified the open ports and running services. I also captured and analyzed the TCP/UDP packet flow using Wireshark, which helped me understand the communication between the Kali machine and the target.